David Israel Vázquez Leal

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Experience

AIST IRCAM (Integrated Research Center for Advanced Manufacturing) **Tokyo, Japan**
Robotics engineer *September 2025 - Present*

- Integrated Jetson and NUC compute stack with LiDAR, stereo cameras, and UR5e for ROS2-based mobile manipulation.
- Designed and implemented an autonomous manipulation pipeline (PCL perception, FSM task control, custom ROS2 interfaces) for robust object picking and placing operations.
- Deployed Nav2 and MoveIt2 to achieve end-to-end autonomous navigation and grasping on a mobile manipulator platform.
- Authored and maintained developer documentation (Sphinx and reStructuredText) and practical usage guides (Markdown), improving reproducibility.

CNRS-AIST JRL (Joint Robotics Laboratory) **Tsukuba, Japan**
Technical Staff *Sep 2024 – Jan 2025*

- Developed a semi-autonomous manipulation pipeline for a mobile manipulator targeting shelf-picking of unknown objects.
- Integrated a UR5e arm with Intel RealSense D435i, ZED Mini, and Jetson Orin Nano via custom ROS2 interfaces.
- Implemented MoveIt-based motion planning and grasp pose estimation using RGB-D perception.
- First author, IEEE/SICE SII 2026: “Development of a semi-autonomous manipulation pipeline for robotic shelf-picking operations”.

Roborregos **Monterrey, Mexico**
Robotics Team Member (Electronics & Software) *Nov 2022 – Jun 2025*

- Led the electronics and navigation subteams as Project Manager.
- Competed in RoboCup@Home 2024 (Eindhoven, Netherlands), contributing to navigation and manipulation.
- Achieved 2nd place at the Mexican Robotics Tournament (@Home) in 2024 and 2025.
- Led the development and embedded architecture of a custom omnidirectional mobile base with integrated LiDAR, Odrive Motors, Jetson and CAN protocol for @Home competition
- Co-first author, IEEE ICRCA 2026: “Software Toolkit for RoboCup@Home: A Modular and Hierarchical Architecture for Service Robots”

VantTec (Self Driving Vehicle development) **Monterrey, Mexico**
Robotics Research Group, Electronics Developer *Apr 2023 – Mar 2024*

- Designed a STM32-based PCB with integrated CAN transceivers to control NEMA 34 stepper motors for steering and braking actuation.

Skills

Programming: C++, Python, C, MATLAB
Robotics: ROS2, Nav2, MoveIt2, UR5e, Robotiq, Jetson, NUC, Gazebo, Behavior Trees
Perception: PCL, RealSense D435i, ZED2/Mini, Orbbec, LiDAR, YOLO
Embedded: STM32, ESP32, FPGA, FreeRTOS, CAN, I2C, SPI, UART
Tools: GitHub, Docker, Doxygen, Sphinx, Markdown, RST, Ubuntu, KiCad/Altium
Languages: Spanish (Native), English (C1)

Education

Instituto Tecnológico de Estudios Superiores de Monterrey (ITESM) **Monterrey, Mexico**
Bachelor of Engineering in Robotics and Digital Systems (96.15/100.00) *August 2021 - July 2025*
Ceneval’s EGEL Excellence Performance Award